Liverpool association for Space engineering & research...

LASER Newsletter

Mach-X team updates



airbrake PCA

Mach-x is firing on all cylinders this week, with every sub-team hard at work! Karl and their team are paving the way for the launch vehicle (AKA the rocket), focusing on CAD design for the Avionics bay, where all the electronics are stored. This leads us nicely into Avionics where Thiruwaran and his team has finalised both their power supply and airbrake board, and is even ready to order and assemble both which is amazing progress! And lastly we have Ben and his CANSAT team have beginning design on their initial schematics and have been paying particular attention on power delivery and CANBus (an electronics and communication protocol for the rocket). Next week plans are being set to start on the rocket's telemetry and main flight computer, so tune back in for the next newsletter when we should have more coverage on those components! That's all for now.

Riddle of the week...
A can fill up a room but take up no space...
What am I?
last weeks answer: Mercury

answer posted next week

WIC
26th October 2026

What's up in the space & engineering world?

Pluto has a new friend? A second hidden dwarf planet in our solar system

October 27th marks a very big day for our solar system, Dr Mathew Holman- an Astrophysicist at Harvard university announced the discovery of a potential new dwarf planet '2026 ED138' orbiting just behind Neptune. This is incredible news as before today we only knew of 5 other dwarf planets.

The discovery was made using data collected by the Vera C. Rubin Observatory. It is estimated that the objects diameter is between 600-1500m wide, and to classify as a dwarf planet and not an asteroid it is estimated the specimen needs to be between 400-900m wide. 2026 ED138 clears that requirement and more, however the real test it meets the gravitational requirements a dwarf planet needs is yet to be confirmed...

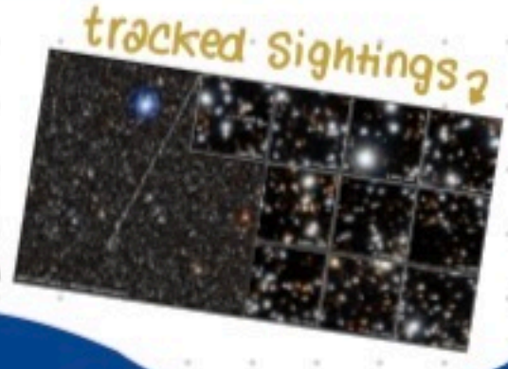
In order to be classified as a dwarf planet the object has to have a mass large enough that it's shaped into a sphere by its own gravitational pull (which is where the 400-900m estimated diameter comes from). It is hoped that NASA's James Webb Telescope will be able to view the object, and using infrared radiation we will be able to narrow down its size and therefore help determine its status as a dwarf planet.

We look forward to seeing more information about '2026 ED138' in upcoming aerospace headlines, as there is so much we still don't know about it. But in the meantime, we can hope that this discovery does lead to a new sibling for Pluto!

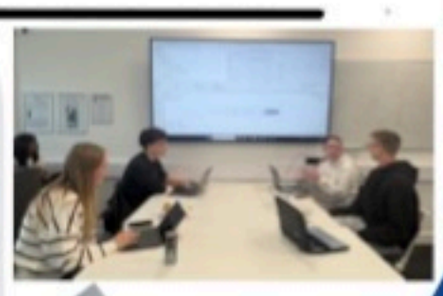
Orbit map ↴



tracked sightings ↴



NRC team updates





This week Unity Rise NRC rocketry team has made a start on its first rocket prototype, using the simulation software 'open rocket'. Right now the team is experimenting with different material options, hoping to find a balance between stability, weight and cost for the different components (such as the fins, body tube and nose cone). On the more electrical side of things Filip has Jac, Kaamini as sub leads focusing on preliminary design for sensors to measure temperature, humidity, altitude and other variables as well as researching options for other electrical components for rocket recovery.

Rover team updates

This week the main focus of the Rover Team is recruitment, with interviews being conducted throughout last week and continuing into this week. Unfortunately all applications have been closed, except software engineer so consider this a last call-out for any aspiring engineers who want to practice their software design skills. The QR code for applications is dropped below, as well as Nathan's email, but I'd like to reiterate that only software engineering applications are being considered. For those who have already applied keep an eye out on your emails, as the final team line up is expected to be confirmed on Sunday. Good luck to anyone waiting to hear back, and we're excited to get a look at the final team later this week.

Email:
sgnallco@liverpool.ac.uk

Application form:

Weekly event updates
Meeting friday 30th October
- personal project info
↳ post meeting notes...
Halloween Sober Social
kahoot Quiz
- General knowledge, Space/engineering, and Halloween theme!

Brought to you by: Olivia Hollidge